

Five Days at Fremont — Part E

AC101 integrated Tesla case · worked in Lecture 5

*Part E of the week-long AC101 Tesla case — worked in Lecture 5. The full fact-vs-fiction note and company background are in the Part A handout (the complete case is on the course site). In short: Tesla is real and the filed figures are real; **every product-costing scenario, internal figure and decision here is hypothetical**; exhibits are labelled REAL, SIMULATED or ESTIMATE. Every question is worked and discussed in the lecture.*

Part E — Friday: the miss — was it the market or the factory?

Friday, 8:05 a.m. The quarter's flash numbers land, and they land badly: operating profit is **\$352m below plan**. Reyes's message beats the coffee to your desk:

"Before this afternoon: was that the market, or the factory? Three bullets. Don't bring me a spreadsheet — bring me who answers for what."

*(The figures in this part are simulated — standards constructed with Tesla's real Q4 2024 as the base, with one stated simplification: cars made = cars sold, so a single volume drives both revenue and cost. The \$9,750m fixed pot is **all** of Tesla's fixed costs — fixed production overhead plus R&D plus SG&A — so the bottom line approximates quarterly operating profit; the magnitude survives a reality check, since Tesla's reported Q4 2024 operating income was \$1,583m (~\$1.6bn) and the case's lands nearby. Reminder: this part is examined at the **final** exam, not the mid-session.)*

Exhibit E-1 lays plan against reality: 495,570 cars delivered against a 500,000 budget; revenue per car a touch soft; materials \$500 a car over standard; labour \$300 under; overheads a whisker adrift. The temptation is to compare actual profit straight against the original budget and start assigning blame. Resist it. The original — **static** — budget was built for 500,000 cars, and the company sold 495,570; comparing across different volumes punishes the factory for the sales team's shortfall. So the first move of any variance analysis: **flex the budget** — restate the plan at the volume actually achieved — and only then compare.

Flexing splits the \$352m miss into two honest pieces. The **sales-volume variance** — the profit lost purely because 4,430 fewer cars were sold, each carrying its budgeted contribution, which is why this variance is **measured on a contribution basis** — and the **flexible-budget variance**, everything that went differently *per car* at the volume actually sold. The first belongs to the demand side of the business; the second to prices and efficiency.

The flexible-budget variance then decomposes line by line (Question 2) — and the decomposition rewrites the story the headline told: the miss is *not* simply "the factory ran badly". Question 2 quantifies which lines actually carry it — and which line, surprisingly, pushes the other way.

But even the materials line is two stories wearing one number, and the tool that separates them is the **standard cost card** (Exhibit E-2). The card sets, in advance, a standard price and a standard quantity per car for each input. Compare actuals against it and the materials line splits into a **price variance** — the factory paid \$50 a unit instead of \$45 — and a **quantity variance** — it used 370 units a car instead of 400. Those two pieces have different owners: purchasing (and the world lithium price) owns the first; the production line owns the second. Net them and you would congratulate or blame the wrong people: the netted "–\$500 a car" was hiding a large price hit almost three-quarters offset by a genuine efficiency gain worth celebrating.

What goes to Reyes at 3 p.m. is therefore not a data dump but a judgement, built on two principles. **Controllability**: hold managers to what they control — blaming the plant for lithium prices or a demand shortfall is unfair, demotivating, and teaches people to pad next year's budget. **Management by exception**: spend

attention only on variances that are *big and controllable* – chase the materials story and the volume question (was it demand, or could we not build enough?), reinforce the labour win, and let the small wobbles wait.

And one last discipline before the memo goes out, supplied by Question 5: a variance is only as trustworthy as the budget behind it, and any control system that watches only profit invites the very trick you discovered on Wednesday.

Assignment questions – Part E

1. From Exhibit E-1, compute the static-budget profit, the flexible-budget profit at actual volume, and actual profit (work in \$m). Split the total variance into the **sales-volume variance, measured on a contribution basis**, and the **flexible-budget variance**. Why must you flex before blaming anyone?
2. Decompose the -\$248m flexible-budget variance line by line at the actual volume of 495,570 cars. Which two numbers would you lead the briefing with?
3. Using Exhibit E-2, split the materials line into its **price** and **quantity** variances, per car and in \$m. Why do the two pieces have different owners, and what management conversation was the netted line hiding?
4. Split the labour line into **rate** and **efficiency** variances – then write the CFO’s three-bullet memo: what drove the miss (numbers, F/U), which variances deserve management attention and who owns each (controllability, management by exception), and one recommendation for next quarter.
5. ► Two challenges before the memo goes out. **(a) Budget quality:** Musk’s public targets are famously aspirational – the “20 million cars by 2030” goal was quietly dropped. What does a variance against a soft or moon-shot budget actually measure, and what does the target-ratchet evidence (Bouwens & Kroos 2011, *Journal of Accounting and Economics*) predict managers will do near year-end? **(b) Gaming:** recalling Part C’s discovery question (Question 5), explain how the plant could have “fixed” this quarter’s miss without selling a single extra car under absorption costing – and why a well-designed control system must watch production alongside profit.
6. Exhibit E-3 lists four further figures that reached your desk after the flash numbers. **(a)** For each figure, say in one line whether it belongs in this quarter’s operating-profit variance, and if it does not, why not. **(b)** Now take the fixed-cost figure – the one that does carry information. Separate the recurring cost base from the one-off charge, state the assumption your treatment depends on, and say how the split changes the management-by-exception verdict on the -\$50m fixed variance.
7. ► Exhibit E-4 opens up the variable-overhead line the briefing waved through as a small wobble. **(a)** Which of the two hour totals is the right activity driver for variable overhead – and why is the other one on the sheet? **(b)** Derive the actual rate per machine-hour. **(c)** Split the variable-overhead line into its spending (rate) and efficiency variances – per car and in \$m – and show they reconcile to the -\$49.6m line. **(d)** In one sentence, what does this line do to the rule “ignore the small variances”?

Exhibits – Part E

Exhibit E-1 – Q4 budget versus actual SIMULATED – standards constructed with Tesla’s real Q4 2024 as the base. Simplifications stated: cars made = cars sold; the fixed pot is ALL fixed costs (fixed production overhead + R&D + SG&A), so the bottom line ≈ quarterly operating profit – a magnitude that survives a reality check against Tesla’s reported Q4 2024 operating income of \$1,583m.

Item	Budget	Actual
Deliveries (cars)	500,000	495,570
Revenue per car delivered*	\$52,000	\$51,900
Direct materials per car	\$18,000	\$18,500
Direct labour per car	\$4,500	\$4,200
Variable overhead per car	\$6,000	\$6,100
Fixed costs (\$m)	9,750	9,800
Derived: variable cost per car	\$28,500	\$28,800
Derived: contribution per car (budget)	\$23,500	—

**Proxy label carried from Part B: total revenue spread over cars delivered, including energy, services and regulatory credits – not an ASP.*

Exhibit E-2 – The standard cost card SIMULATED – labelled, like the rest of this part's figures.

Input	Standard	= per car	Actual this quarter
Direct materials	400 units @ \$45	\$18,000	370 units @ \$50
Direct labour	60 hours @ \$75	\$4,500	50 hours @ \$84

Exhibit E-3 – Four further figures on the analyst's desk *Mixed sources – each row labelled REAL or SIMULATED. These figures reached your desk after the flash numbers; decide which, if any, belong in the quarter's operating-profit variance. The \$9,800m actual fixed cost in Exhibit E-1 is unchanged – row 4 is a component of it, not an addition to it.*

#	Item	Figure	Label
1	Tesla operating profit, prior quarter (Q3 2024)	~\$2,700m	REAL – Tesla Q3 2024
2	Fremont paint-shop line (installed 2023), remaining book value	~\$600m	SIMULATED – illustrative
3	Regulatory-credit revenue booked in Q4	~\$690m	REAL – Tesla Q4 2024
4	One-off restructuring/severance charge sitting <i>inside</i> the \$9,800m actual fixed costs (not in the budget)	\$80m	SIMULATED – illustrative

Exhibit E-4 – Variable-overhead standard card and Q4 actuals SIMULATED – *an illustrative internal driver split, consistent with the rest of this part's figures; variable overhead per car (\$6,000 budget, \$6,100 actual) is carried unchanged from Exhibit E-1. The actual rate per machine-hour is Question 7.*

Item	Standard	Actual
Variable-overhead rate per machine-hour	\$1,000	[?]
Machine-hours per car	6.0	5.0
<i>Variable overhead per car</i>	<i>\$6,000</i>	<i>\$6,100</i>

Quarter totals (actual)	Figure
Total variable overhead	\$3,023m
Total machine-hours	2.478m
Total direct-labour hours	24.78m

Friday, 4 p.m.

The briefing slide you finally show Reyes has one sentence at the top: *"It depends which cost you mean – and here is which one you should mean, for each decision on your desk."*

Underneath it, the week:

- **Monday:** the published accounts say Tesla as a whole made money — \$17.45bn gross, \$7.08bn operating — but they contain no cost of a *Model Y*. Cost object first; then the classification that matches the question.
- **Tuesday:** for “*does the price war sink us?*”, the estimated fixed/variable split — a ~\$22,648 contribution per car against a ~\$4.3bn quarterly fixed pile — gives Tesla a margin of safety of roughly 58%: actual deliveries run about 2.4 times the COGS break-even volume. The cut survives if volume holds; SG&A and R&D still sit on top.
- **Wednesday:** for “*which model makes the money?*”, the allocated full cost — done properly, with ABC — shows the volume cars carrying ~86% of the profit while the premium badges supply the garnish.
- **Thursday:** for “*make or buy?*”, only the avoidable and opportunity costs count — and the same \$800m can be irrelevant to one question and decisive for the next.
- **Friday:** for “*who answers for the miss?*”, the standard cost — flexed to actual volume, split into price and quantity, judged by controllability.

Use the wrong cost for the day’s decision and you hand a CFO a confidently wrong answer. Use the right one and the numbers do what management accounting is for: they narrow the choice — and then judgement finishes it.

Selected references woven through the case: Anderson, Banker & Janakiraman (2003), “Are selling, general, and administrative costs ‘sticky’?”, Journal of Accounting Research; Roychowdhury (2006), “Earnings management through real activities manipulation”, Journal of Accounting and Economics; Bouwens & Kroos (2011), “Target ratcheting and effort reduction”, Journal of Accounting and Economics.